

Exploring Mission Status and Rocket Performance

Global Space Launches

1. Project Overview

This dataset contains information about global space launches conducted by various space agencies and private aerospace companies across different countries. The dataset includes launch details, rocket status, mission outcomes, launch locations, company origin, launch dates, and mission costs.

The purpose of this project is to analyse trends in global space missions, identify successful launch patterns, compare private and government organizations, and visualize launch activities across countries and years.

Objectives

Analyse Launch Frequency
Evaluate Mission Success Rates
Compare Private vs State-Run Organizations
Geographic Analysis
Time-Based Trend Analysis
Dataset Description

The dataset records individual space launch events from multiple organizations such as:

- SpaceX
- Roscosmos
- ISRO
- ULA
- CASIC
- Rocket Lab
- Arianespace
- Blue Origin
- Northrop
- JAXA

Each row represents one launch mission.

3. Data Source

Dataset collected from publicly available space launch records and mission databases.

Possible sources include:

- Spaceflight databases

- Aerospace mission records
- Public launch history archives

4. Dataset Columns Description

Column Name	Description
Company Name	Name of the organization conducting the launch
Location	Launch site location
Detail	Rocket and mission details
Status Rocket	Operational status of rocket
Rocket	Cost/value of rocket mission
Status Mission	Launch result (Success/Failure/Partial Failure)
Country of Launch	Country where launch occurred
Companys Country of Origin	Country where company belongs
Private or State Run	Indicates whether organization is private (P) or state-run (S)
DateTime	Full launch date and time
Year	Launch year
Month	Launch month
Day	Launch day
Date	Launch date
Time	Launch time

Data Sources

Dataset Source Exploring Mission Status and Rocket Performance

- **Timeline:** 2020 – 2021
- **Data Size:** 4324 rows and 15 to 16 columns

Prescriptive Analytics (What Should Be Done?)

Prescriptive analytics recommends actions for better operational decisions.

Recommendations

- Increase investment in reusable rockets
- Improve quality checks for high-failure rocket families
- Expand private-public partnerships
- Optimize launch schedules during peak success periods

- Enhance risk monitoring for experimental launches

Diagnostic Analytics (Why Did It Happen?)

Diagnostic analytics identifies reasons behind patterns and failures.

Examples

- Why China had the highest launch frequency
- Why SpaceX achieved higher success rates
- Why certain rocket families had repeated failures
- Relationship between rocket status and mission outcome
- Impact of private vs state-run organizations

Data Transformation

Created calculated columns:

- Success/Failure Category
- Launch Year
- Month Name
- Private vs State Classification
- Rocket Status Grouping

Global Space Launches Dataset – Attribute / Column Feature Details

Column Name	Description	Data Type	Example
Company Name	Name of the organization/company that conducted the launch mission	Text	SpaceX, CASIC, ISRO
Location	Launch site or launch pad where the rocket was launched	Text	LC-39A, Kennedy Space Centre, Florida, USA
Detail	Rocket name and mission/payload details	Text	Falcon 9 Block 5 Starlink V1 L9 & Black Sky
Status Rocket	Operational status of the rocket	Text / Categorical	Status Active, Status Retired

Rocket	Cost of the rocket launch (mostly in million USD)	Decimal Number	50, 145, 29.75
Status Mission	Result/status of the mission	Text / Categorical	Success, Failure, Partial Failure
Country of Launch	Country where the launch took place	Text	USA, China, India
Company's Country of Origin	Country where the company originates from	Text	USA, Russia, China
Private or State Run	Indicates whether the company is Private or Government/State-owned	Text / Categorical	P = Private, S = State Run
DateTime	Complete launch date and time with timezone	DateTime	2020-08-07 05:12:00+00:00
Year	Launch year extracted from DateTime	Whole Number	2020
Month	Launch month extracted from DateTime	Whole Number	8
Day	Launch day extracted from DateTime	Whole Number	7
Date	Launch date only	Date	07-08-2020
Time	Launch time only	Time	05:12

Project Domains

1. Space & Aerospace Analytics

Best and most direct domain for this dataset.

You can analyze:

- Rocket launches by year/month
- Successful vs failed missions
- Top space agencies
- Private vs government launches
- Launch frequency trends

- Country-wise launch performance

Relevant entities:

- SpaceX
 - Roscosmos
 - NASA
 - JAXA
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2. Business Intelligence & Data Analytics

Excellent for:

- Power BI dashboards
- KPI reporting
- Trend analysis
- Interactive visualizations
- Executive reporting

Possible KPIs:

- Total launches
 - Mission success rate
 - Top launch countries
 - Most active company
 - Average rocket cost
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3. Transportation & Engineering Analytics

Since rockets are advanced transportation systems, the dataset can be used for:

- Engineering performance analysis
- Vehicle reliability
- Launch scheduling
- Operational efficiency

Relevant entity:

- Aerospace Engineering
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4. Government vs Private Sector Analysis

Your column:

Private or State Run

makes this ideal for:

- Public vs private space race analysis
- Commercial space industry growth
- Investment trend studies

Example comparisons:

- SpaceX vs Roscosmos
 - USA vs China launch ecosystem
 - Commercial launch success rates
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5. Time Series & Predictive Analytics

Since your dataset has:

- Date
- Time
- Year
- Month
- Day

You can build:

- Forecasting models
 - Launch prediction analysis
 - Seasonal trends
 - Mission growth analysis
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Best Final Project Titles

Here are strong project title ideas:

1. **Global Space Launch Mission Analysis**
 2. **Space Industry Analytics Dashboard**
 3. **Worldwide Rocket Launch Intelligence**
 4. **Private vs Government Space Missions Analysis**
 5. **Global Aerospace Launch Trends**
 6. **Rocket Mission Success Analytics**
 7. **Space Exploration Data Visualization Using Power BI**
 8. **International Space Launch Performance Dashboard**
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Best Tools for This Dataset

- Microsoft Power BI
 - Microsoft Excel
 - Tableau
 - Python
 - SQL
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Suggested Dashboard Pages

Overview Dashboard

- Total launches
- Total countries
- Success rate
- Active companies

Company Analysis

- Launches by company
- Success/failure ratio
- Rocket cost comparison

Country Analysis

- Country-wise launches
- Top launch centers
- Geographic insights

Time Analysis

- Yearly trend
- Monthly trend
- Launch timeline

Mission Analysis

- Mission outcomes
- Rocket status
- Most used rockets

Data Cleaning Process

The following cleaning operations were performed in Power Query:

Removed Errors

- Removed invalid or null records.

Fixed Encoding Issues

- Corrected special character problems in location names and mission details.

Standardized Country Names

Examples:

- Usa → USA
- Isreal → Israel

Converted Data Types

- DateTime converted to Date/Time datatype
- Rocket column converted to Decimal Number
- Year, Month, Day converted to Whole Number

Handled Missing Values

- Blank rocket cost values replaced with null or average values where necessary.

Created Additional Columns

- Extracted Year, Month, Day from DateTime
- Split Date and Time columns

6. Business Problems Solved

The project answers the following analytical questions:

1. Which company conducted the highest number of launches?
2. Which country launched the most missions?
3. What is the overall mission success rate?
4. Which rockets had the highest failure rates?
5. Comparison between private and government organizations
6. Year-wise launch trends
7. Monthly launch distribution
8. Most active launch locations
9. Cost analysis of launch missions

10. Country-wise mission success comparison

7. Key Insights

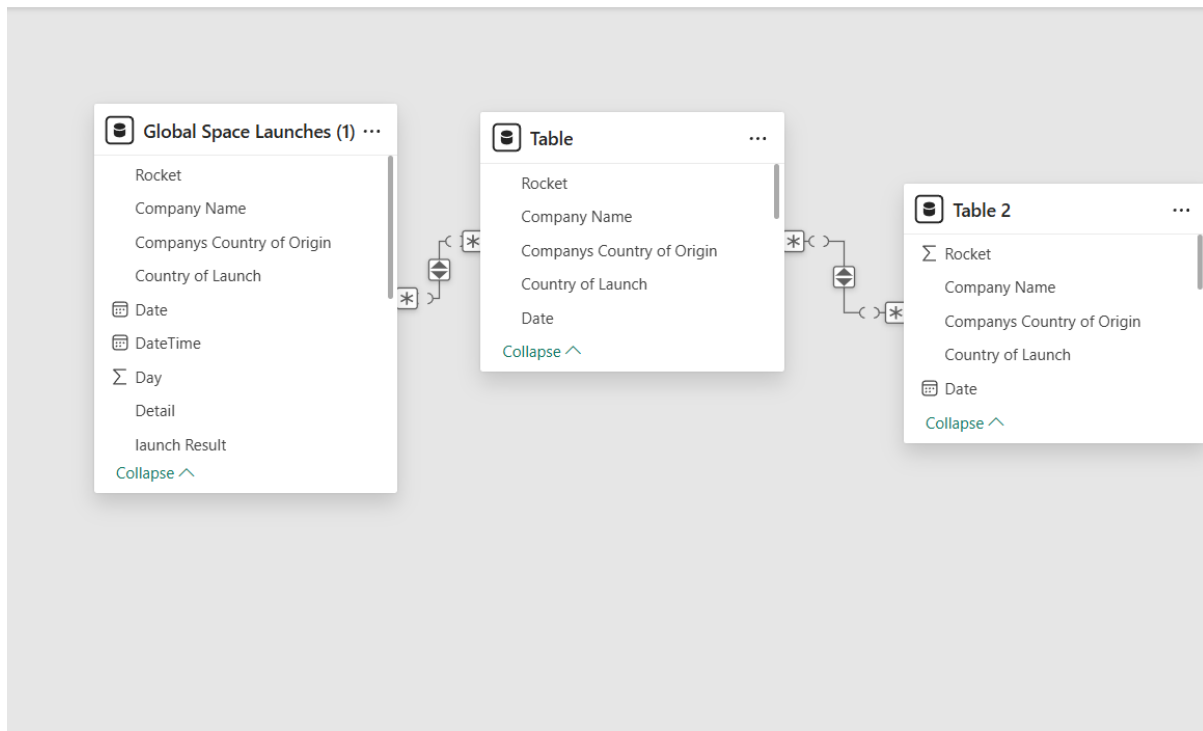
- SpaceX became one of the most active launch providers.
 - China conducted a high number of state-run launches.
 - Most missions had successful outcomes.
 - Private companies showed rapid growth in launch frequency.
 - USA, China, and Russia dominated global launch activities.
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8. Dashboard Features

The Power BI dashboard includes:

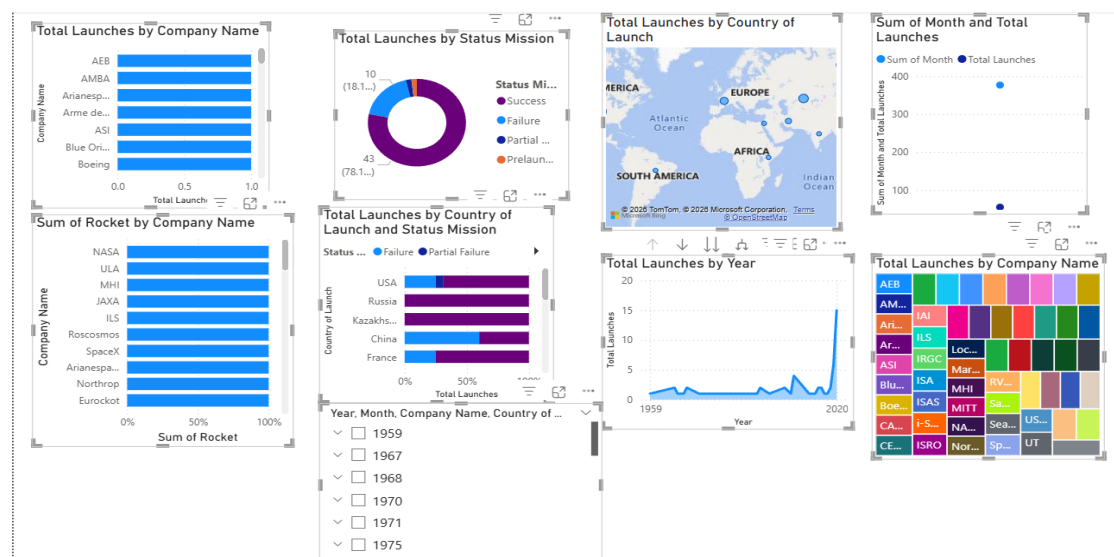
- KPI Cards
 - Total Launches
 - Successful Missions
 - Failed Missions
 - Success Rate
- Charts
 - Year-wise launches
 - Country-wise launches
 - Company performance
 - Mission status analysis
 - Private vs State-run comparison
- Filters/Slicers
 - Year
 - Country
 - Company
 - Mission Status

Data modelling



Dashboard Overview

The dashboard contains multiple analytical sections to provide business insights into space launch operations.



Key Insights (SMART + Analytics Approach)

1. Launch Success Dominates Overall Missions

- Around **78% of launches** were successful, while failure and partial failure missions form a much smaller share.

- This indicates a **high operational reliability trend** in global space missions.
 - **SMART Insight:**
 - **Specific:** Focus on reducing failure missions.
 - **Measurable:** Reduce failure rate by 5–10%.
 - **Achievable:** Through better testing and launch simulations.
 - **Relevant:** Improves mission efficiency and cost savings.
 - **Time-Bound:** Target improvement within the next 2 years.
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2. USA and Russia Lead Global Launch Activity

- The dashboard shows the **USA and Russia** contribute the highest number of launches globally.
 - Emerging participation is visible from countries like **China, France, and Kazakhstan**.
 - **Business Insight:** These countries dominate due to advanced space infrastructure and higher investments in aerospace programs.
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3. Launches Increased Significantly Over Time

- The yearly trend chart shows a **sharp rise in launches near 2020**.
 - This suggests:
 - Growth of private aerospace companies.
 - Increased satellite deployments.
 - Expansion of commercial space programs.
 - **Predictive Insight:** Launch demand is expected to continue growing due to satellite internet, defense, and research missions.
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4. Certain Companies Drive Most Space Missions

- Companies/agencies such as:
 - Boeing
 - ArianeSpace
 - NASA
 - SpaceX
 - Roscosmoscontribute significantly to total launches.
 - **Strategic Insight:** The market is highly concentrated among a few dominant aerospace organizations.
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5. SpaceX and Private Sector Growth are Emerging Trends

- SpaceX appears among major contributors, showing the transition from government-led to **commercial space operations**.
 - **Diagnostic Insight:** Cost-efficient reusable rockets are changing the aerospace industry model.
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6. Rocket Contributions are Highly Distributed

- Multiple rocket providers contribute nearly equally in some areas.
 - This indicates:
 - Strong competition.
 - Technology diversification.
 - Reduced dependency on a single launch provider.
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7. Geographic Spread of Launches

- The map visualization highlights launches concentrated in:
 - North America
 - Europe
 - Russia
 - Asia
 - **Geospatial Insight:** Space programs are heavily linked to technologically advanced economies.
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SMART Recommendations

Objective	Recommendation	KPI
Improve mission success	Enhance testing & AI monitoring	Failure rate reduction
Increase launch efficiency	Use reusable rocket systems	Cost per launch
Expand global collaboration	Partner with emerging countries	International missions
Forecast future demand	Apply predictive analytics models	Launch growth trend
Optimize resource allocation	Analyze company-wise performance	ROI per mission

Types of Analytics Used in This Dashboard

1. Descriptive Analytics

Purpose: Understand “What happened?”

Used in:

- Total launches by company
- Total launches by year
- Country-wise launches
- Mission status distribution

Outcome: Summarizes historical launch performance.

2. Diagnostic Analytics

Purpose: Understand “Why did it happen?”

Used in:

- Mission success vs failure analysis
- Country vs mission status comparison
- Company performance breakdown

Outcome: Identifies reasons behind mission outcomes and operational trends.

3. Predictive Analytics

Purpose: Forecast “What is likely to happen?”

Used in:

- Yearly launch trend growth
- Increasing private aerospace participation
- Future launch demand estimation

Outcome: Predicts future growth in space missions and commercial launches.

4. Prescriptive Analytics

Purpose: Recommend “What should be done?”

Used in:

- Improving mission reliability
- Resource optimization
- Strategic planning for launch operations

Outcome: Suggests actions for better performance and reduced failure rates.

5. Geospatial Analytics

Purpose: Analyze location-based patterns.

Used in:

- Country launch map visualization

9. Tools Used

Tool	Purpose
Power BI	Dashboard and visualization
Power Query	Data cleaning and transformation
Excel/CSV	Raw dataset storage

10. Conclusion

This project provides insights into global space launch activities and demonstrates data cleaning, transformation, visualization, and analytical reporting skills using Power BI and Power Query.

The analysis helps understand:

- Space industry growth
 - Mission success patterns
 - Country dominance in space exploration
 - Role of private aerospace companies
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